



ENGINEERING MATHEMATICS-II

Kavyashree

Department of Science and Humanities



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FOURIER TRANSFORMS

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Problem 4: Solve the integral equation $\int_0^{\infty} f(x) \sin(xt) dx = \begin{cases} 1, & 0 < t < 1 \\ 2, & 1 < t < 2 \\ 0, & \text{otherwise} \end{cases}$

Solution: We know that $F_s(\omega) = \int_0^{\infty} f(t) \sin \omega t dt = \mathcal{F}_s[f(x)]$

$$\text{Given that, } \int_0^{\infty} f(x) \sin x t dt = \begin{cases} 1, & 0 < t < 1 \\ 2, & 1 < t < 2 \\ 0, & \text{otherwise} \end{cases} = \mathcal{F}_s[f(x)]$$

$$\text{i.e., } F_s(\omega) = \begin{cases} 1, & 0 < \omega < 1 \\ 2, & 1 < \omega < 2 \\ 0, & \text{otherwise} \end{cases} = \mathcal{F}[f(t)]. \text{ Therefore, in order to find } f(t) \text{ we need to find inverse transform of } F_s(\omega)$$

$$\text{Thus, } f(t) = \frac{2}{\pi} \int_0^{\infty} F_s(\omega) \sin \omega t d\omega$$

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Fourier Sine Transforms :



$$= \frac{2}{\pi} \left[\int_0^1 \sin \omega t d\omega + \int_1^2 2 \sin \omega t d\omega \right]$$

$$= \frac{2}{\pi} \left[\left\{ \frac{-\cos \omega t}{t} \right\}_0^1 + \left\{ \frac{-2 \cos \omega t}{t} \right\}_1^2 \right]$$

$$= \frac{2}{\pi} \left[\left\{ \frac{-(\cos t - 1)}{t} \right\} + \left\{ \frac{-2(\cos 2t - \cos t)}{t} \right\} \right]$$

$$f(t) = \frac{2}{\pi} \left[\left\{ \frac{1 - 2 \cos 2t + \cos t}{t} \right\} \right]$$

Fourier Sine Transforms:



Problem 5: Find Fourier Sine Transform of e^{-at} ($a > 0$) and hence show that $\int_0^{\infty} \frac{x \sin kx}{x^2 + a^2} dx = \frac{\pi}{2} e^{-ak}$

Solution: $\mathcal{F}_s[f(t)] = \int_0^{\infty} f(t) \sin \omega t dt \Rightarrow \mathcal{F}_s[e^{-at}] = \int_0^{\infty} e^{-at} \sin \omega t dt$

$$= \left[\frac{e^{-at}}{a^2 + \omega^2} \{-a \sin \omega t - \omega \cos \omega t\} \right]_0^{\infty}$$

$$= \frac{\omega}{a^2 + \omega^2} = F_s(\omega)$$

$$f(t) = \frac{2}{\pi} \int_0^{\infty} F_s(\omega) \sin \omega t d\omega$$

$$= \frac{2}{\pi} \int_0^{\infty} \frac{\omega}{a^2 + \omega^2} \sin \omega t d\omega$$

Fourier Sine Transforms:



$$\Rightarrow e^{-at} = \frac{2}{\pi} \int_0^{\infty} \frac{\omega \sin \omega t}{a^2 + \omega^2} d\omega$$

$$t = k \Rightarrow e^{-ak} \frac{\pi}{2} = \int_0^{\infty} \frac{\omega \sin \omega k}{a^2 + \omega^2} d\omega;$$

$$\omega \rightarrow x \Rightarrow \boxed{\int_0^{\infty} \frac{x \sin kx}{a^2 + x^2} dx = \frac{\pi}{2} e^{-ak}}$$

Fourier Sine Transforms:



Fourier sine transform of derivatives:

Let $f(t)$ and $f'(t)$ be continuous on $[0, \infty)$. Let $f(t) \rightarrow 0$ and $f'(t) \rightarrow 0$ as $t \rightarrow \infty$ and $f''(t)$ be piecewise continuous on every subinterval $[0, l]$. Then,

$$\mathcal{F}_s[f''(t)] = -\omega^2 F_s(f(t)) + \omega f(0)$$

Proof: By definition we have,

$$\begin{aligned}\mathcal{F}_s[f''(t)] &= \int_0^{\infty} f''(t) \sin \omega t dt \\ &= \left[\sin \omega t f'(t) \right]_0^{\infty} - \int_0^{\infty} \omega \cos \omega t f'(t) dt\end{aligned}$$

Fourier Sine Transforms:



$$= 0 - \omega \int_0^{\infty} \cos \omega t f'(t) dt = -\omega \left\{ [\cos \omega t f(t)]_0^{\infty} + \omega \int_0^{\infty} \sin \omega t f(t) dt \right\}$$

$$= -\omega \left\{ 0 - f(0) + \omega \int_0^{\infty} \sin \omega t f(t) dt \right\}$$

$$\mathcal{F}_s[f''(t)] = -\omega^2 F_s(f(t)) + \omega f(0)$$



THANK YOU

Kavyashree

Department of Science and Humanities

kavyashree@pes.edu